

Temporal Change in Modeled Connectivity Among
Southern African Cheetah (*Acinonyx jubatus jubatus*)
Range Cores, 2012–2024

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Abstract

The southern African Cheetah (*Acinonyx jubatus jubatus*), has had their range decreased drastically in the past decades. This has only been exacerbated in the past decades due to both natural and anthropogenic reasons, leaving their current extant range majority outside protected areas (Weise et al., 2017b). This leaves barriers like ranchers, roads and other human development in areas where cheetahs will be roaming (Marker et al., 2008). Models looking at habitat suitability have been done in the past, but very little predictive modeling of the movement network permeability. One 2026 KAZA (Kavango-Zambezi Transfrontier Conservation Area) multi-species connectivity study excluded the cheetah outright (Sousa et al., 2026). A similar study was completed for the Asiatic cheetah (*Acinonyx jubatus venaticus*), identifying potential connectivity through the Iranian landscape (Moqanaki & Cushman, 2017). This study attempts to identify core areas within southern Africa using population data from the Dryad dataset accompanying Weise et al. (2017b) (Weise et al., 2017a), model connectivity between them using a multivariate analysis, then see how that connectivity has changed between 2012 and 2024. The modeled connection cost changed only slightly overall. Out of the 253 core pairs, 161 became somewhat easier and 92 became harder. Outside the core areas, only about 23% of the landscape that is most important for cheetah movement was protected. This is lower than the 31.26% of protected land across the wider study area. Protection was higher only in the very most connected places. Of the 45 links selected by the model, 32 were kept as the main priority links. The other 13 are still possible connections, but need more field data before they can be treated as priorities.

Keywords: landscape connectivity; circuit theory; *Acinonyx jubatus*; protected area coverage; monitoring prioritisation; southern Africa

1 Introduction

The southern African cheetah has had their range substantially reduced, with their current range estimated to be on majority unprotected land. Cheetah range on protected land is governed by centralized park-management initiatives (Buk et al., 2018). Unprotected land poses a different type of challenge, with landowners, local communities and governmental entities involved. This paper looks at landscape connectivity change between 2012 and 2024. The core range of cheetahs was fixed at the 2010-2016 baseline, so modeled changes in permeability reflected landscape changes rather than changes in the core range itself.

Previous research in the region has predominantly focused on habitat suitability, a topic that is already comprehensively addressed in the existing literature. The objective of my study was not to develop an additional suitability model, but rather to identify and evaluate candidate spatial “pinch points” (i.e., constricted linkages) among core areas. The study most comparable to the present work is a recent multi-species connectivity assessment; however, that analysis explicitly excluded cheetah due to insufficient data. Data limitations were therefore explicitly considered and mitigated to the extent possible in this study, and they provide both the primary justification for the existence of this paper and its principal constraint.

Two primary connectivity approaches are commonly used to model these kinds of frameworks: least-cost connectivity and circuit-theoretic connectivity (Adriaensen et al., 2003; McRae et al., 2008). The least-cost approach links core areas by selecting the route with the lowest cumulative “cost,” meaning the path of least resistance between cores, based on weight values assigned to each multivariate resistor. In contrast, the circuit-theoretic approach represents the landscape as a resistor network, sends a “current” through it, and highlights redundancy across the broader landscape, relying on the same multivariate resistors. A key limitation of both methods is that you must first assign a resistance “score” or “weight” to every cell (1 km in my case). Zeller et al. (2012) demonstrated that even reasonable alternative assumptions about resistance values can influence results more strongly

than the choice of connectivity method itself. For that reason, rather than treating one resistance surface as definitive, I varied the weights along two axes. The first changed the balance among the anthropogenic pressures, comparing my primary weighting against an equal weighting and an infrastructure-emphasis weighting. The second changed how much the surface leaned on vegetation relative to anthropogenic pressure and terrain, at 10, 20 and 40 percent. The vegetation-balanced surface combined with the documented fence treatment carries the primary results, and the remaining combinations are reported as sensitivity rather than as alternative answers. A link is only described as stable here if it survives them. This helps to minimize bias from the selection of resistor weights.

Temporal-change hypothesis

Temporal change in connectivity is expected to be strongest among a minority of links, and the links that are most important to the network are expected to experience the greatest loss of change between 2012 and 2024. The importance of a link is measured by its "betweenness" in the network, which is the number of shortest paths that end up passing through it. A link with a high betweenness is therefore more important to the network than a link with a low betweenness. This was compared between all 253 core pairs. The expectation is that the most important links will be the ones most sensitive to change over time.

Protected-area-coverage hypothesis

High-value connectivity areas are those with the highest connectivity values. Cells that are part of the protected areas are expected to have higher connectivity values than cells in the unprotected areas. The proportion of the cheetah's range that is currently protected is expected to be a significant factor in maintaining connectivity. The protected-area used was the most recent available data and was not reconstructed at each historical date. Areas with higher levels of protection are anticipated to have lower rates of connectivity loss compared to areas with less protection.

Link-stability hypothesis

The stability of a link is defined as its ability to maintain connectivity under varying conditions. Links that are more stable are expected to be less affected by changes in the landscape over time. Some links may be more sensitive to changes in the landscape due to their location or the surrounding land use (i.e., a link that is surrounded by agricultural land may be more sensitive to changes in land use than a link that is surrounded by protected areas). The expectation is that links that are more stable will experience less change in connectivity over time compared to links that are less stable.

Throughout, potential structural connectivity refers to modelled landscape permeability between fixed range cores. It is not evidence of cheetah movement, dispersal, or gene flow. Where a location is described as a pinch point, this means a location of concentrated modelled current or a link-level bottleneck, not an observed crossing.

2 Methods

2.1 Study area and analysis extent

The study area was defined as the southern African cheetah range (Weise et al., 2017b) plus a buffered analysis extent. All layers were aligned to a fixed equal-area projection and a 1 km snap grid, so every raster was directly comparable cell for cell. Cores were derived from the historical confirmed-range baseline and held constant across 2012, 2016, 2020, and 2024. The final network consisted of 23 cores and 45 selected neighboring core pairs. I used a 1-km working resolution because it represented a compromise between the rougher 10-km occurrence baseline and the finer environmental covariates, which ranged from 30 m to 1 km. This resolution was fine enough to identify meaningful points in the landscape while avoiding the implication of a level of accuracy that the occurrence evidence alone could not provide.

2.2 Range cores

I defined a core area as a contiguous patch of cells that were confirmed presence, with a density of ≥ 0.51 cheetahs per 100 km^2 . The final network consisted of 23 cores, which were held constant across all four temporal snapshots to avoid inventing core movement in the absence of comparable range reassessment.

I represented these 23 core areas as fixed nodes across all four temporal snapshots (2012, 2016, 2020 and 2024). I took each core area and calculated the Euclidean distance to the nearest neighbor. This yielded 45 straight-line links. These straight-line links defined the lines used to get the least cost paths based on the added resistance surface rasters.

2.3 Covariates and their temporal treatment

The predicted temporal resistance values came from the continuous rasters (human development from GHSL, nighttime lights from VIIRS, and VCF (vegetation) from MODIS). Categorical land cover was used for masking and description only, because annual categorical features can switch classes between years for reasons unrelated to overall landscape change.

I also used the static covariates (roads, livestock, terrain, hydrography, fences, protected areas, and core definitions). The static covariates did not contribute *temporal* signal to the resistance surface; however, they did contribute *spatial* signal to the resistance surface. The static covariates were used to describe the landscape and to provide context for the dynamic covariates.

2.4 Resistance scenarios

Resistance values were assigned to represent the relative difficulty of movement through each landscape feature. The terrain component was derived from slope, with steeper slopes assigned higher resistance values. The final balanced map had a weight of 60% for human

pressure, 20% for vegetation, and 20% for terrain. Within human pressure, built-up land, roads, and livestock each received 30%, while nighttime lights received 10%. Available fence data was incorporated into the resistance model, with fence cells increasing the combined resistance by a multiplier of 1 if there is no fence, or 25 if there is a fence cell. Fences did not incur a full block to cheetah movement as they are not complete barriers (Cozzi et al., 2013). I treated slopes near 15 percent as easiest for movement and assigned more resistance above or below that point. This had to be done carefully however, given cheetahs do have an affinity for some sloped terrains (Moqanaki & Cushman, 2017). I compared alternative vegetation weights because vegetation cover can influence cheetah habitat use (Verschueren et al., 2024). Different weighting schemes allowed me to test how strongly the modeled resistance surfaces depended on the vegetation weight (Zeller et al., 2012). The final resistance surface was used to model least-cost paths and Circuitscape current flow between core areas.

2.5 Connectivity modelling

For each core, I ran a least-cost path, in which ArcGIS calculated the accumulated cost of moving through every reachable cell. From there, the lowest accumulated cost was used to draw a path from each connected core back to the starting core. Doing it this way meant that I would be reusing one cost calculation for several destinations, which gives the same number of pairwise paths with fewer repeat calculations. I then used Circuitscape 5.17.1 (Anantharaman et al., 2020) to spread out all possible routes between each pair of cores. This was done for all 253 core pairs, which were solved in each of the four years which came to a total of 1,012 pair comparisons. Fig. 4 shows the current output for 2024.

2.6 Network structure

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What was measured: edge betweenness counts how often a link lies on the shortest network route between pairs of cores. A high value means the link helps hold many parts of the modeled network together. The value was

calculated on the 45-link network and divided by the 253 possible core pairs. How links were chosen: the network combined each core’s three nearest neighbors with a minimum spanning tree, which is the smallest set of links needed to keep the core network connected. Why rankings are uncertain: allowing two, three, four, or five nearest neighbors created 31, 45, 61, or 77 links, and the highest-ranked links changed. Link 17–24 was the only bridge under the three-neighbor choice; removing a bridge separates the network into two pieces. What was dropped: PC and dPC (Saura & Pascual-Hortal, 2007) require an assumed relationship between distance and dispersal. No local dispersal telemetry was available, so those more detailed-looking numbers were not calculated. Cite Keeley et al. (2021) when explaining that the metric should match the research question.]

2.7 Temporal comparison

[TODO: PLAIN-LANGUAGE WRITING GUIDE: compare the same core pairs, model assumptions, and 1-km grid in 2012, 2016, 2020, and 2024. This keeps the comparison consistent so the modeled differences come from the input layers that changed through time. Define absolute change as later value minus earlier value. Define percent change as that difference divided by the earlier value, multiplied by 100.]

2.8 Protected-area coverage

[TODO: PLAIN-LANGUAGE WRITING GUIDE. Question: what share of the modeled connecting landscape outside the cores falls inside protected areas? Data preparation: use the August 2026 WDPA release (UNEP-WCMC & IUCN, 2026). Keep terrestrial and coastal protected-area polygons, remove proposed areas, points, and marine-only areas, and merge overlaps. This left 2,301 records in the study extent. Comparison area: 31.26 percent of the usable outside-core area was protected. Cells with zero current were excluded because they could never enter a high-current group. Why no p-value: the answer changed from significant to not significant when the assumed spatial block size changed. Report the descriptive

percentages and note that the same general pattern appeared in all four years and under several core-buffer sizes.]

2.9 Stability and uncertainty

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What "stable" means here: a link had to pass three checks. First, both alternative weightings had to keep at least 80 percent of the route within 5 km of the main route; 34 of 45 passed. Second, the fence scenarios could not substantially change the link; this removed two more from the group. Third, the direction of change had to stay the same under all vegetation weightings. The two links that failed this last check had already been removed. The final groups were 32 main-priority links and 13 uncertain links. Important distinction: the 80-percent-within-5-km rule is not the cell-by-cell agreement test proposed in the original protocol, even though both use the number 80 percent. What was not completed: alternative core definitions, 2-km and 5-km grids, and a cell-by-cell agreement map. Separate location uncertainty from cost uncertainty: 49 of 180 route-and-year comparisons had less than 80 percent overlap within 1 km, including 21 main-priority links. Link 17–24 had no overlap in 2020. This does not mean the connection was unimportant; it means the exact mapped route was uncertain.]

2.10 Independent plausibility assessment

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What was planned: compare the model with separate Limpopo tracking data and Namibia camera-trap data. What was actually done: compare early and late occurrence records from the same archive used to create the cores. Because the data are not independent, call this a limited plausibility check rather than validation. Numbers available: after thinning to one record per 10-km cell, 2010–2012 had 413 cells and 2013–2016 had 450. Their overlap score was 0.239. There are no records after 2016, so this check says nothing about 2020 or 2024. Decision still needed: either complete the planned outside-data check or clearly state that it was not completed.]

2.11 Monitoring prioritisation

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What was planned: combine six measurements into one score for every 10-km cell and test three sets of weights. What was built instead: a link-level table that keeps each kind of evidence separate. Its columns describe how important the link is to the network, how its modeled cost changed, how stable its mapped route was, whether fences affected it, and how much of it is protected. Why this is more honest: combining everything into one score would require deciding, for example, how much route instability is equal to a change in protection. No evidence-based exchange rate was available. State clearly that no combined monitoring score was calculated.] There is a cheeta

2.12 Software and reproducibility

[TODO: PLAIN-LANGUAGE WRITING GUIDE: list the software without explaining every technical setting in the main text. Verified versions are ArcGIS Pro 3.7.2 (build 1901), ArcGIS Python 3.13.13, Julia 1.12.7, and Circuitscape.jl 5.17.1; see Supplementary Table 5. Google Earth Engine supplied and aligned the satellite data. Most ArcGIS processing used Python scripts rather than ModelBuilder. Add the final archive link and state which source datasets cannot legally be redistributed.]

3 Results

3.1 Connectivity change, 2012–2024

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What happened: I compared all 253 possible pairs of cores. Overall, the modeled cost of moving between cores changed only a little from 2012 to 2024. The middle change was -1.224 percent. A negative value means the model became slightly easier to move through; it does not mean that cheetahs actually

moved more. Of the 253 pairs, 92 became harder and 161 became easier. The largest increase was +13.1 percent for the link between cores 33 and 38. What the map adds: outside the cores, the places with the highest modeled current were mostly in the same locations through time. Their overlap scores were 0.91 to 0.93, where 1 would mean perfect overlap. What not to claim: do not call this an overall connectivity loss, observed movement, or a country-by-country result. The analysis did not assign each core pair to a country. For H1, report that 18 of the 45 selected links became harder and 27 became easier, but that change was not meaningfully related to baseline betweenness ($\rho \approx 0.01$). Therefore H1 was mixed rather than supported. Use Figure 3.]

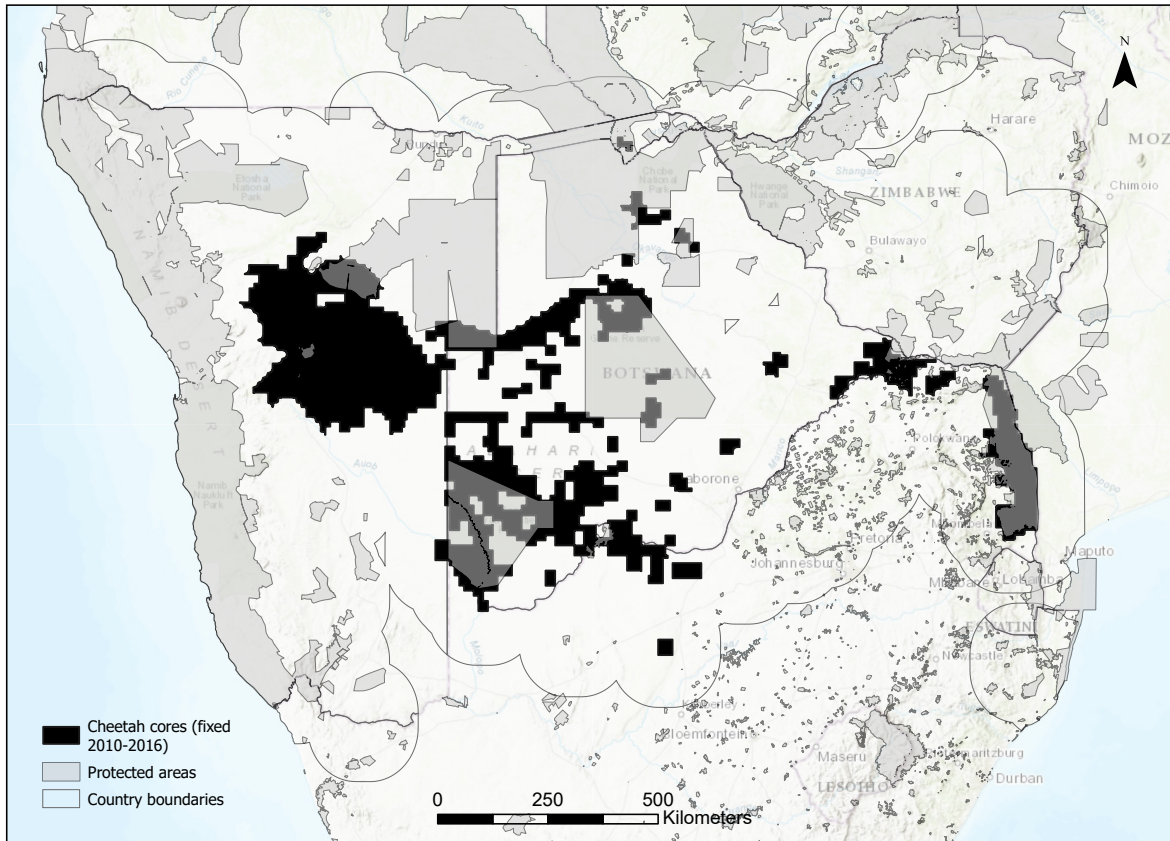


Figure 1: Study extent, fixed historical range cores, and core groups used to define the modeled connectivity network. Core groups are an analysis construct and do not imply separate populations.

3.2 Priority links and sensitivity checks

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What happened: 32 of the 45 selected links passed the tested sensitivity screens and remained in the main priority group. Network importance was ranked separately using betweenness. The other 13 links were kept as uncertain links where more field information would be useful. The weighting test kept 34 links, and the fence test removed two more. The vegetation test did not remove any additional links from the remaining group. What it means: the main group is not a small handful of links; it contains most of the network. That only partly matches the original expectation that priorities would be limited to a small subset. What not to claim: these links are modeled connections between cores, not confirmed animal travel routes. Use Table 2.]

[TODO: PLAIN-LANGUAGE WRITING GUIDE FOR THE SENSITIVITY TESTS. Main lesson: the total modeled cost of a connection was usually fairly stable, but the exact line taken by the least-cost path was less stable. Of 180 route-and-year comparisons, 49 had less than 80 percent overlap within 1 km. This affected 21 of the 32 priority links. The route between cores 17 and 24 had no overlap in 2020. Vegetation assumptions: 43 of 45 links kept the same direction of change under all three vegetation weightings. However, the size of the change grew as vegetation was given more weight: the middle absolute change increased from about 3.6 to 9.3 percent. Network-design assumption: changing how many neighbors each core could connect to changed the number of links and changed which links ranked highest. A bridge is a link whose removal splits the network into separate pieces; link 17–24 was the only bridge when three nearest neighbors were used. What not to claim: there is no completed cell-by-cell agreement map, and no single ranking should be presented as certain. State that H3 was only partly evaluated: a stable group existed, but it contained a majority of the selected links and the full planned sensitivity grid was not completed.]

[TODO: PLAIN-LANGUAGE WRITING GUIDE FOR THE OCCURRENCE CHECK. What happened: after thinning records to one 10-km cell, the early period had 413 occupied cells and the later period had 450. Their overlap score was 0.239, showing that the two

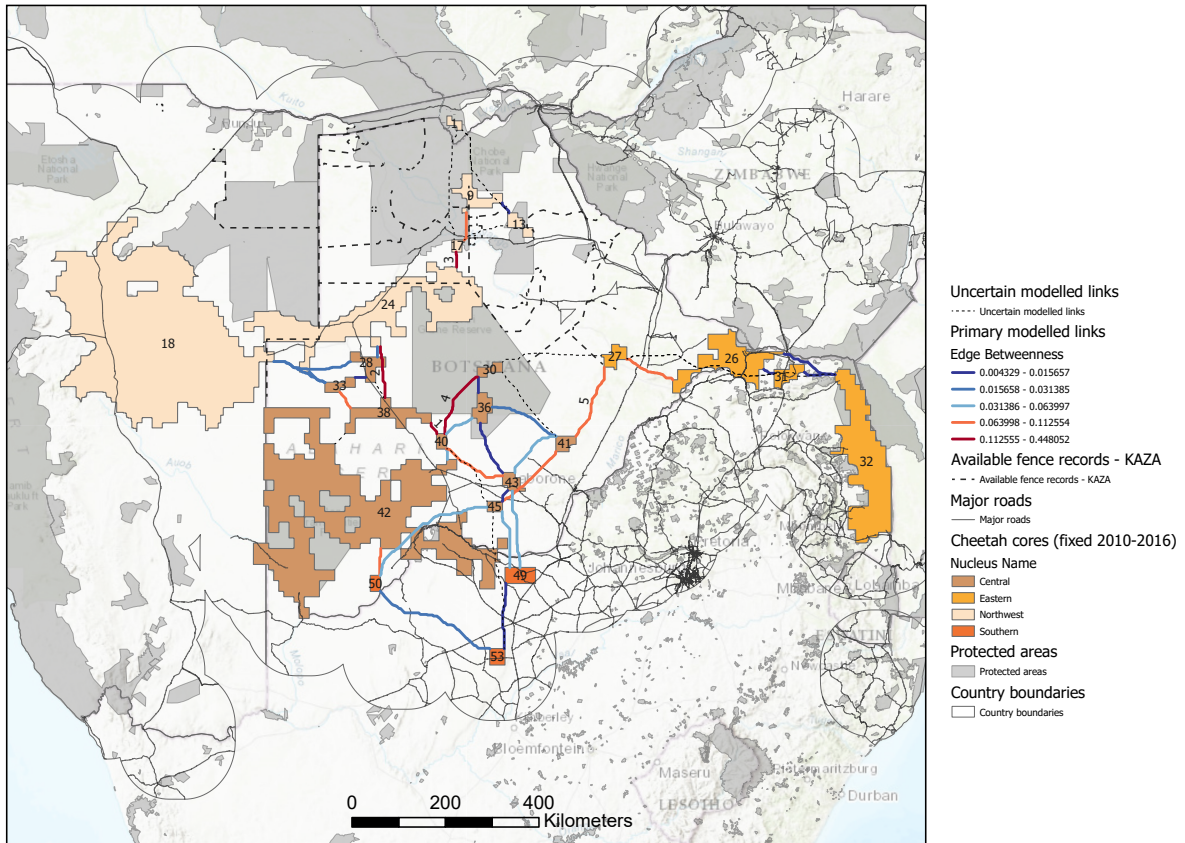


Figure 2: Modeled core-to-core links with transportation and available fence context. Lines represent network links used to define least-cost analyses; they are not observed animal paths.

sets were quite different. Later records occurred in slightly higher modeled resistance, and a larger share fell inside the fixed cores. Almost no records were close to the priority paths: zero early cells and three later cells. Why this is a weak check: the same occurrence archive helped create the cores, so this is not an independent test. There are also no occurrence records after 2016, so it cannot check the 2020 or 2024 results. Report it honestly as a limited comparison, not as validation.]

[TODO: Protocol reporting only: the planned cell-level monitoring index was not built and H4 was not tested. Explain this once in Appendix A, not as a repeated priority-link result.]

3.3 Protected-area coverage of priority connectivity

[TODO: PLAIN-LANGUAGE WRITING GUIDE. Why the cores were removed: the model forces current into and out of the cores, so the cores automatically receive very high current. Including them would make protection look better without telling us how well the land between cores is protected. What happened outside the cores: 31.26 percent of the comparison area is protected. Only 23.27 percent of the area above the 80th current percentile was protected in 2024. Coverage rose to 25.54 percent above the 90th percentile, 33.45 percent above the 95th, and 44.63 percent above the 99th. In simple terms, the very strongest concentrations often reached protected areas, while the broader connecting landscape around them was less protected. What not to claim: do not report a significance p-value because it changed when the size of the spatial blocks changed. Do not call modeled current observed movement. Use Table 3. State that H2 was not supported under its full rule because the direction changed across current thresholds and the significance result was not stable.]

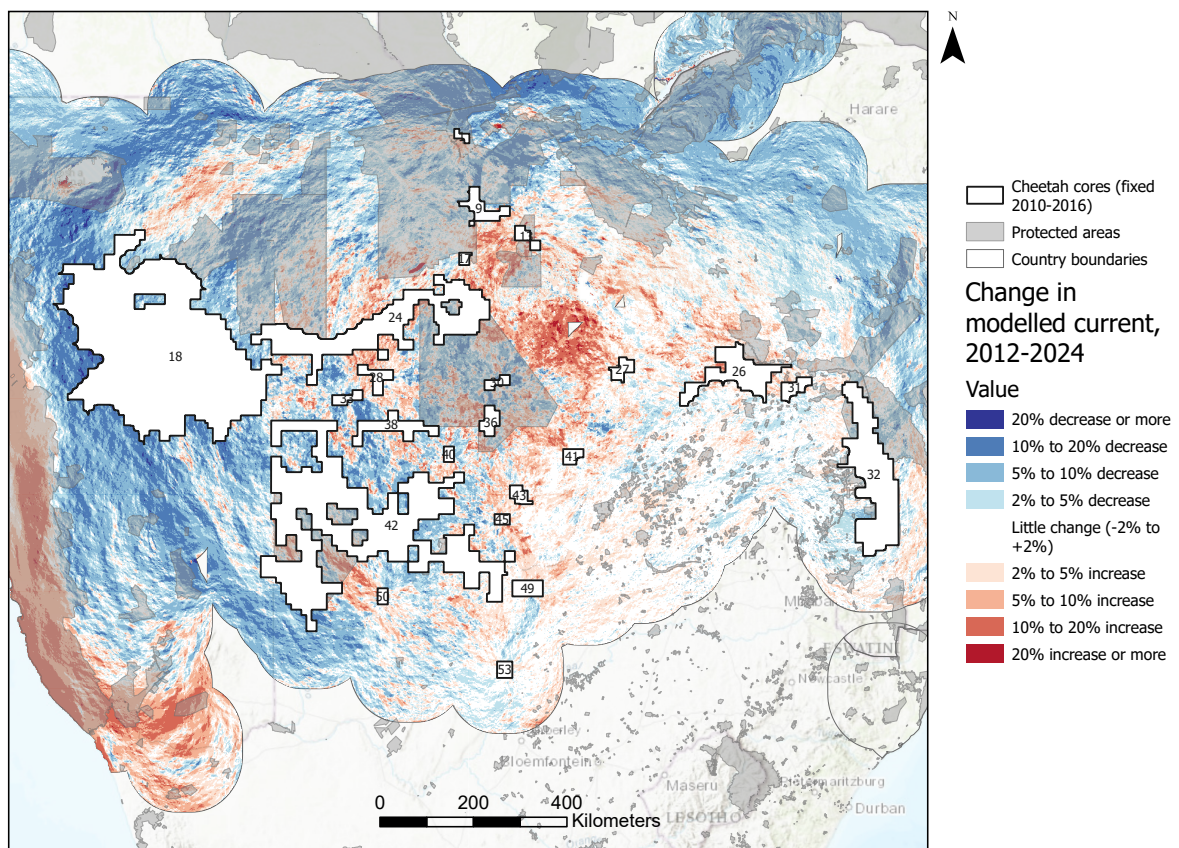


Figure 3: Modeled current change between 2012 and 2024.

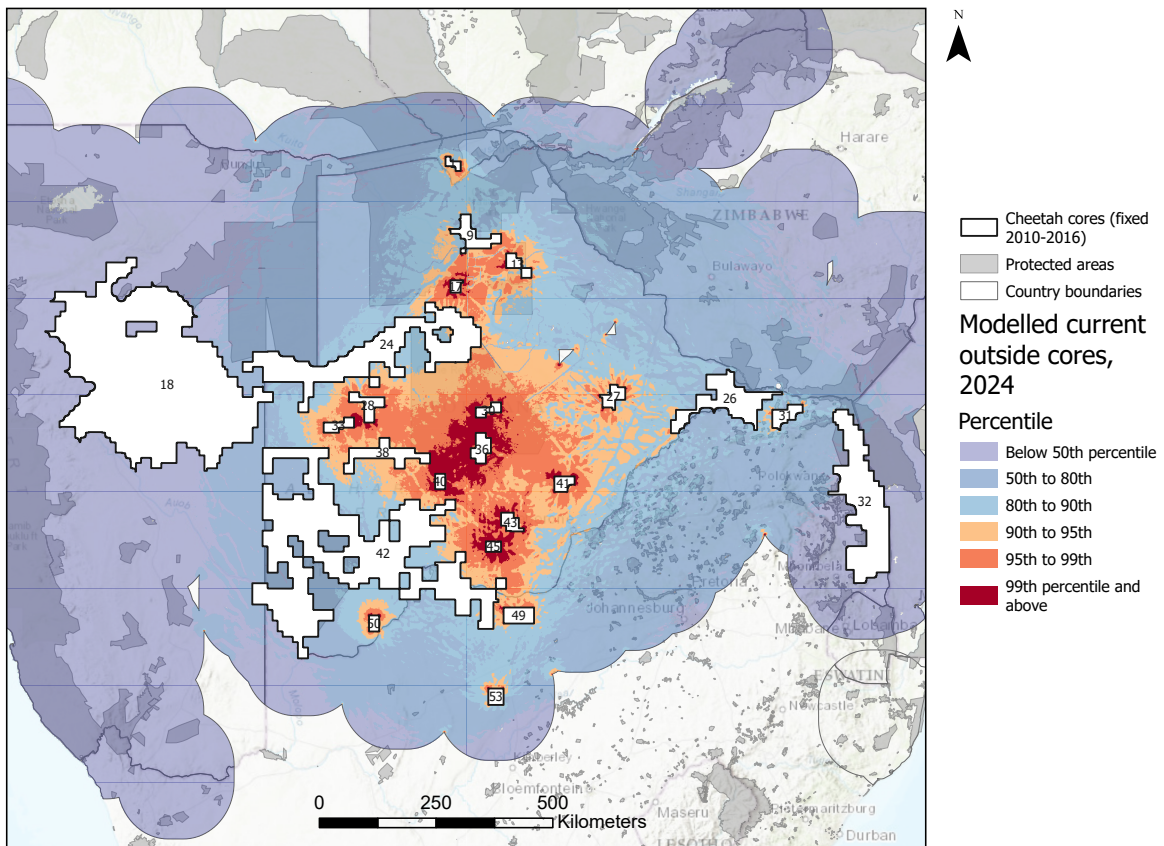


Figure 4: Modeled current concentration for 2024, with the fixed core network shown for orientation.

Table 1: Covariates, native resolution, source years, and static/dynamic treatment.

Covariate	Source	Treatment	Temporal role / notes
Built-up fraction	Global Human Settlement Layer (GHSL)	Dynamic	2010, 2015, 2020, 2025 products matched to snapshots
Nighttime radiance	Visible Infrared Imaging Radiometer Suite (VIIRS), VNL V2	Dynamic	2013, 2016, 2020, 2024 products
Tree cover	Moderate Resolution Imaging Spectroradiometer (MODIS), MOD44B	Dynamic	Annual snapshot fields
Non-tree vegetation	Moderate Resolution Imaging Spectroradiometer (MODIS), MOD44B	Dynamic	Annual snapshot fields
Bare cover	Moderate Resolution Imaging Spectroradiometer (MODIS), MOD44B	Dynamic	Annual snapshot fields
Road density	Global Roads Inventory Project, version 4 (GRIP4)	Static	Spatial pressure/context
Livestock density	Gridded Livestock of the World, version 4 (GLW4)	Static	Combined cattle, goats, and sheep exposure
Terrain / slope	Shuttle Radar Topography Mission (SRTM)	Static	Slope-derived resistance
Hydrography	Hydrological data and maps based on Shuttle Elevation Derivatives at multiple Scales (HydroSHEDS)	Static	Water/context mask
Veterinary fences	World Wide Fund for Nature (WWF), Kavango–Zambezi Transfrontier Conservation Area (KAZA); Kruger references	Scenario	Sensitivity/context, not assumed impassable

Note: Product shorthand retained in the source names is defined at first use: VNL = VIIRS nighttime lights; V2 = product version 2; MOD44B = MODIS vegetation-continuous-fields product.

Table 2: Ten highest-ranked modeled priority links shown from the 32-link primary-priority set.

From	To	Rank	Betweenness	Type	Distance (km)	Change (%)
38	40	1	0.409	KNN+MST	120.6	+4.78
24	38	2	0.316	KNN	202.3	+7.44
17	24	3	0.300	KNN+MST	208.0	-5.78
30	40	4	0.238	KNN	169.2	+3.55
27	41	5	0.103	KNN+MST	196.8	-6.45
41	45	6	0.090	KNN	183.6	-5.50
33	38	7	0.079	KNN	108.1	+13.12
9	17	8	0.079	KNN+MST	109.0	-5.26
26	27	9	0.079	KNN+MST	221.8	-5.18
40	43	10	0.077	KNN	159.6	-1.56

Notes: The table shows the ten highest-ranked members of the 32-link primary-priority set; links are modeled structural network relationships, not observed animal routes. KNN = k -nearest-neighbor link; MST = minimum spanning tree link. “KNN+MST” identifies links retained by both network-construction rules.

Table 3: Protected-area coverage of modeled high-current classes.

Year	Current percentile	Cells	Protected (%)	Background (%)	Difference (pp)
2012	80	418,295	24.23	31.26	-7.03
2012	90	209,148	26.31	31.26	-4.95
2012	95	104,574	35.57	31.26	+4.31
2012	99	20,915	44.17	31.26	+12.92
2024	80	418,292	23.27	31.26	-7.99
2024	90	209,146	25.54	31.26	-5.72
2024	95	104,573	33.45	31.26	+2.19
2024	99	20,915	44.63	31.26	+13.37

[TODO: Final note: values use the outside-core, positive-current domain. Percentage-point differences are relative to the matched background. Do not report a permutation p -value because inference was sensitive to the assumed spatial block size.]

Table 4: Overview of completed modeled-connectivity results.

Metric	Value	Interpretation
Final modeled-current snapshots	4 (2012, 2016, 2020, 2024)	253/253 focal-region pairs connected each year
Fixed density cores	23	Historical baseline, fixed across snapshots
Selected neighboring links	45	Core-pair comparison set
Primary-priority links	32	Stable under weight/fence screens
All-pair median effective-resistance change	-1.224%	2012–2024
Selected-link median change	-1.557%	2012–2024
Primary-priority median change	-2.618%	2012–2024
Largest all-pair increase	+13.119%	Link 33–38
Largest all-pair decrease	-8.243%	Link 43–45

4 Discussion

4.1 Connectivity change and comparison with previous work

[TODO: PLAIN-LANGUAGE GUIDE: begin with the simplest overall result: the modeled ease of moving between cores changed only slightly overall, but some individual links changed more than others. Immediately state that the model describes landscape structure and does not show where cheetahs actually moved.]

[TODO: PLAIN-LANGUAGE GUIDE: describe where modeled connections became easier or harder and which landscape changes occurred in the same places. Say "occurred in the same places," not "caused." The model found mixed change rather than a broad decline. Do not describe connectivity by country because core pairs were not assigned to individual countries.]

[TODO: PLAIN-LANGUAGE GUIDE: compare the general mapping approach with the elephant and multi-species studies (Brennan et al., 2020; Naidoo et al., 2024) and with the fence study (Osipova et al., 2018). Explain that similar mapping tools can be used for different species, but the numerical resistance values cannot simply be copied because each

species responds differently to roads, vegetation, fences, and people.]

4.2 Protected areas and conservation context

[TODO: PLAIN-LANGUAGE GUIDE: the strongest modeled concentrations often pass through protected areas, but the wider land connecting those areas is less protected. Explain that conservation therefore also depends on farms, communal lands, and other land outside formal parks. This does not prove that protected areas failed. Also note that parks are often located on land with lower agricultural value, which may partly shape the pattern.]

4.3 Priority links and field-survey needs

[TODO: PLAIN-LANGUAGE GUIDE: links that stayed important under several assumptions are the strongest candidates for conservation attention. Links that changed substantially between assumptions are useful places for field surveys because better local data could change the result. Agreement between scenarios is not complete proof because the scenarios share many of the same input layers.]

[TODO: PLAIN-LANGUAGE GUIDE: there is no completed score that ranks individual survey cells. Instead, explain how an organization could use the 13 uncertain links to choose general survey areas, then adjust the order based on access, cost, safety, and local knowledge. Link 17–24 is especially useful for checking because it is the only link holding two parts of the three-neighbor network together, yet its mapped route was highly unstable and had no overlap in 2020.]

5 Limitations

[TODO: SENTENCE 1: Say that the model shows where movement may be easier or harder, but it does not prove where cheetahs actually traveled. SENTENCE 2: Say that the cores and several landscape layers were kept constant through time, allowing a fair comparison but

limiting what could change between years. SENTENCE 3: Say that the resistance values came from published research rather than GPS movement data and that differences in data coverage, rainfall, and source years add uncertainty. SENTENCE 4: Say that the mapped paths are broad planning areas rather than exact travel routes and that the results should be tested before use at another scale or in another region.]

6 Conclusion

[TODO: PLAIN-LANGUAGE GUIDE: use only three or four sentences. First, say that modeled connectivity changed only slightly overall and that individual links changed in both directions. Second, say that the strongest concentrations often reached protected areas but the wider connecting landscape was less protected. Third, state that 32 links were retained as main priorities and 13 were identified as places needing more field information. End by reminding the reader that these are modeled landscape connections, not observed cheetah routes. Do not add new results here.]

Data availability

[TODO: PLAIN-LANGUAGE GUIDE: add the final repository or archive link and DOI. State that the scripts and outputs created by the project are available there. Explain that raw protected-area data must be downloaded from Protected Planet under its license and that cheetah occurrence locations can only be shared at the detail allowed by their source.]

Acknowledgements

[TODO: PLAIN-LANGUAGE GUIDE: list the department, advisor, data providers, and any organization or person who reviewed the project or helped interpret priorities.]

Conflicts of interest

I declare no conflicts of interest.

References

- Adriaensen, F., Chardon, J. P., De Blust, G., Swinnen, E., Villalba, S., Gulinck, H., & Matthysen, E. (2003). The application of ‘least-cost’ modelling as a functional landscape model. *Landscape and Urban Planning*, 64(4), 233–247. [https://doi.org/10.1016/S0169-2046\(02\)00242-6](https://doi.org/10.1016/S0169-2046(02)00242-6)
- Anantharaman, R., Hall, K., Shah, V. B., & Edelman, A. (2020). Circuitscape in Julia: High performance connectivity modelling to support conservation decisions. *Proceedings of the JuliaCon Conferences*, 1(1), 58. <https://doi.org/10.21105/jcon.00058>
- Brennan, A., Beytell, P., Aschenborn, O., Du Preez, P., Funston, P. J., Hanssen, L., Kilian, J. W., Stuart-Hill, G., Taylor, R. D., & Naidoo, R. (2020). Characterizing multispecies connectivity across a transfrontier conservation landscape. *Journal of Applied Ecology*, 57(9), 1700–1710. <https://doi.org/10.1111/1365-2664.13716>
- Buk, K. G., van der Merwe, V. C., Marnewick, K., & Funston, P. J. (2018). Conservation of severely fragmented populations: Lessons from the transformation of uncoordinated reintroductions of cheetahs (*Acinonyx jubatus*) into a managed metapopulation with self-sustained growth. *Biodiversity and Conservation*, 27(13), 3393–3423. <https://doi.org/10.1007/s10531-018-1606-y>
- Cozzi, G., Broekhuis, F., McNutt, J. W., & Schmid, B. (2013). Comparison of the effects of artificial and natural barriers on large african carnivores: Implications for interspecific relationships and connectivity. *Journal of Animal Ecology*, 82(3), 707–715. <https://doi.org/10.1111/1365-2656.12039>

- Keeley, A. T. H., Beier, P., & Jenness, J. S. (2021). Connectivity metrics for conservation planning and monitoring. *Biological Conservation*, 255, 109008. <https://doi.org/10.1016/j.biocon.2021.109008>
- Marker, L. L., Dickman, A. J., Mills, M. G. L., Jeo, R. M., & Macdonald, D. W. (2008). Spatial ecology of cheetahs on north-central namibian farmlands. *Journal of Zoology*, 274(3), 226–238. <https://doi.org/10.1111/j.1469-7998.2007.00375.x>
- McRae, B. H., Dickson, B. G., Keitt, T. H., & Shah, V. B. (2008). Using circuit theory to model connectivity in ecology, evolution, and conservation. *Ecology*, 89(10), 2712–2724. <https://doi.org/10.1890/07-1861.1>
- Moqanaki, E. M., & Cushman, S. A. (2017). All roads lead to iran: Predicting landscape connectivity of the last stronghold for the critically endangered asiatic cheetah. *Animal Conservation*, 20(1), 29–41. <https://doi.org/10.1111/acv.12281>
- Naidoo, R., Beytell, P., Brennan, A., Carter, J., Carter, K. D., Chamaillé-Jammes, S., Chilambe, B., Hoare, R., Iiyambo, N., Jooste, D., Karidozo, M., Kilian, J. W., Madhlamoto, D., Madiri, T. H., McCulloch, G., Monks, N., Mudimba, I., Ngwenya, N., Nyambe, N., ... Songhurst, A. (2024). Landscape connectivity for african elephants in the world’s largest transfrontier conservation area: A collaborative, multi-scalar assessment. *Journal of Applied Ecology*, 61(10), 2483–2496. <https://doi.org/10.1111/1365-2664.14746>
- Osipova, L., Okello, M. M., Njumbi, S. J., Ngene, S., Western, D., Hayward, M. W., & Balkenhol, N. (2018). Fencing solves human-wildlife conflict locally but shifts problems elsewhere: A case study using functional connectivity modelling of the african elephant. *Journal of Applied Ecology*, 55(6), 2673–2684. <https://doi.org/10.1111/1365-2664.13246>
- Saura, S., & Pascual-Hortal, L. (2007). A new habitat availability index to integrate connectivity in landscape conservation planning: Comparison with existing indices and

- application to a case study. *Landscape and Urban Planning*, 83(2-3), 91–103. <https://doi.org/10.1016/j.landurbplan.2007.03.005>
- Sousa, L. L., Kotze, R., Fanikiso, L., Young-Overton, K., Overton, J., Stevens, X., Lines, R., Wijers, M., Kesch, K., Bauer, D. T., Seymour-Smith, J., Sibanda, A., Parry, R., Fitt, M., Burger, K., Seyoka, S., McNutt, J. W., Stein, A. B., Pelayo-Malet, V., ... Loveridge, A. J. (2026). Multispecies transboundary landscape connectivity in the KAZA TFCA. *Scientific Reports*, 16(1), 24330. <https://doi.org/10.1038/s41598-026-58440-y>
- UNEP-WCMC & IUCN. (2026, August). *Protected planet: The world database on protected areas (WDPA)*. Cambridge, UK. <https://doi.org/10.34892/6fwd-af11>
- Verschueren, S., Hofmann, T., Briers-Louw, W. D., Kakove, M., Leirs, H., Bauer, H., Marker, L., & Cristescu, B. (2024). A diminished large carnivore guild with contrasting species-habitat associations persists outside national parks in namibia’s central-eastern landscape. *Biological Conservation*, 297, 110741. <https://doi.org/10.1016/j.biocon.2024.110741>
- Weise, F. J., Vijay, V., Jacobson, A. P., Schoonover, R. F., Groom, R. J., Horgan, J., Keeping, D., Klein, R., Marnewick, K., Maude, G., Melzheimer, J., Mills, G., van der Merwe, V., van der Meer, E., van Vuuren, R. J., Wachter, B., & Pimm, S. L. (2017a). *Data from: The distribution and numbers of cheetah (Acinonyx jubatus) in southern africa*. Dryad Digital Repository. <https://doi.org/10.5061/dryad.7n4h4>
- Weise, F. J., Vijay, V., Jacobson, A. P., Schoonover, R. F., Groom, R. J., Horgan, J., Keeping, D., Klein, R., Marnewick, K., Maude, G., Melzheimer, J., Mills, G., van der Merwe, V., van der Meer, E., van Vuuren, R. J., Wachter, B., & Pimm, S. L. (2017b). The distribution and numbers of cheetah (acinonyx jubatus) in southern africa. *PeerJ*, 5, e4096. <https://doi.org/10.7717/peerj.4096>

Zeller, K. A., McGarigal, K., & Whiteley, A. R. (2012). Estimating landscape resistance to movement: A review. *Landscape Ecology*, 27(6), 777–797. <https://doi.org/10.1007/s10980-012-9737-0>

A Analysis protocol as predeclared

[TODO: PLAIN-LANGUAGE GUIDE FOR THE PROTOCOL APPENDIX. The saved protocol is tagged `protocol-v1.0` in the `cheetah-connectivity` repository at commit `d7de8675a074dfbffc0c7c856bb5645e1e13867d`, dated 18 August 2026. However, the file itself still says version 0.1 draft. Do not change the old file. Explain that the Git tag marked the saved version even though its internal label was not updated. Then list, in simple terms, what changed between the plan and the final analysis: the core definition, sensitivity tests, definition of a stable link, scenario names, network measurements, protected-area comparison, occurrence check, monitoring index that was not built, and 2030 scenarios that were added later. Include the objective and hypothesis outcomes: O1 was completed; O2 was narrowed; O3 was replaced rather than completed; H1 was mixed; H2 did not pass its full rule; H3 was only partly tested and did not match the “limited subset” prediction; and H4 was not tested.]

B Supplementary material

[TODO: PLAIN-LANGUAGE GUIDE: the tables below preserve details that are useful for checking the analysis but are too long for the main paper. They cover which source year matched each modeled year, strongly related predictors, overlap of high-current areas, software versions, vegetation assumptions, route stability, and the 2030 scenarios. Do not add tables for different map resolutions or a cell-level monitoring score because those analyses were not completed.]

[TODO: PLAIN-LANGUAGE DATA CHECK: every value in these tables should come

Table 5: [TODO: Final caption: software environment used for the completed analysis.]

Component	Version	Verification
ArcGIS Pro	3.7.2 (build 1901)	<code>arcpy.GetInstallInfo()</code>
ArcGIS Python	3.13.13	ArcGIS Pro environment
Julia	1.12.7	<code>julia -version</code>
Circuitscape.jl	5.17.1	Julia package environment
Google Earth Engine	Cloud service	Acquisition and harmonisation

Table 6: [TODO: Final caption: temporal matching of dynamic covariates to the four analysis snapshots.]

Covariate	Source	Treatment	2012 snapshot	2016 snapshot	2020 snapshot	2024 snapshot
Built-up fraction	GHSL	Dynamic	2010 (−2)	2015 (−1)	2020 (0)	2025 (+1)
Nighttime radiance	VIIRS VNL V2	Dynamic	2013 (+1)	2016 (0)	2020 (0)	2024 (0)
Tree cover	MODIS MOD44B C6.1	Dynamic	2012 (0)	2016 (0)	2020 (0)	2024 (0)
Non-tree vegetation	MODIS MOD44B C6.1	Dynamic	2012 (0)	2016 (0)	2020 (0)	2024 (0)
Bare ground	MODIS MOD44B C6.1	Dynamic	2012 (0)	2016 (0)	2020 (0)	2024 (0)
Land cover	MODIS MCD12Q1 C6.1	Context only	2012 (0)	2016 (0)	2020 (0)	2024 (0)

[TODO: Final note: values in parentheses are source-year minus nominal-year offsets.
Define GHSL, VIIRS, VNL, and MODIS here.]

Table 7: [TODO: Final caption: predictor pairs with absolute Spearman correlation of at least 0.75.]

Predictor pair	Spearman ρ
Tree cover–bare ground	−0.877
Non-tree vegetation–bare ground	−0.836
Cattle–goats	0.774

[TODO: Final note: report only pairs with $|\rho| \geq 0.75$ from the archived 2024 predictor screen.]

Table 8: [TODO: Final caption: pairwise overlap of outside-core high-current cells among successive snapshots and endpoints.]

Earlier year	Later year	Earlier cells	Later cells	Intersection	Union	Jaccard
2012	2016	209,148	209,150	200,069	218,229	0.917
2016	2020	209,150	209,147	201,179	217,118	0.927
2020	2024	209,147	209,146	199,881	218,412	0.915
2012	2024	209,148	209,146	199,511	218,783	0.912

[TODO: Final note: cells are the outside-core top decile of positive modeled current in each snapshot.]

Table 9: [TODO: Final caption: conditional 2030 scenario changes in selected-link least-cost path cost relative to 2024.]

Scenario	Links	Median change (%)	Minimum (%)	Maximum (%)
Low growth	45	0.00	0.00	0.00
Continuation	45	−3.36	−8.53	+6.06
High development	45	−6.94	−17.12	+8.65

[TODO: Final note: change is relative to the exact 2024 low-growth baseline; these are conditional development scenarios, not forecasts.]

Table 10: [TODO: Final caption: sensitivity of 2012–2024 selected-link cost change to vegetation weighting.]

Vegetation weighting	Links	Median absolute change (%)
Low	45	3.57
Balanced	45	6.27
High	45	9.28

[TODO: Final note: 43 of 45 links retained the same 2012–2024 change direction under all three vegetation weightings.]

Table 11: [TODO: Final caption: route-location sensitivity under the corrected vegetation inputs.]

Link set	Links	Route-years	Route-years below 80%	Affected links
All selected links	45	180	49	29
Primary-priority links	32	128	34	21

[TODO: Final note: the threshold requires at least 80 percent reciprocal route overlap within 1 km; it is a sensitivity screen, not a biological cutoff.]

from a CSV in `tables/source_data/`. Never copy a number from a screenshot or estimate it from a map legend.]

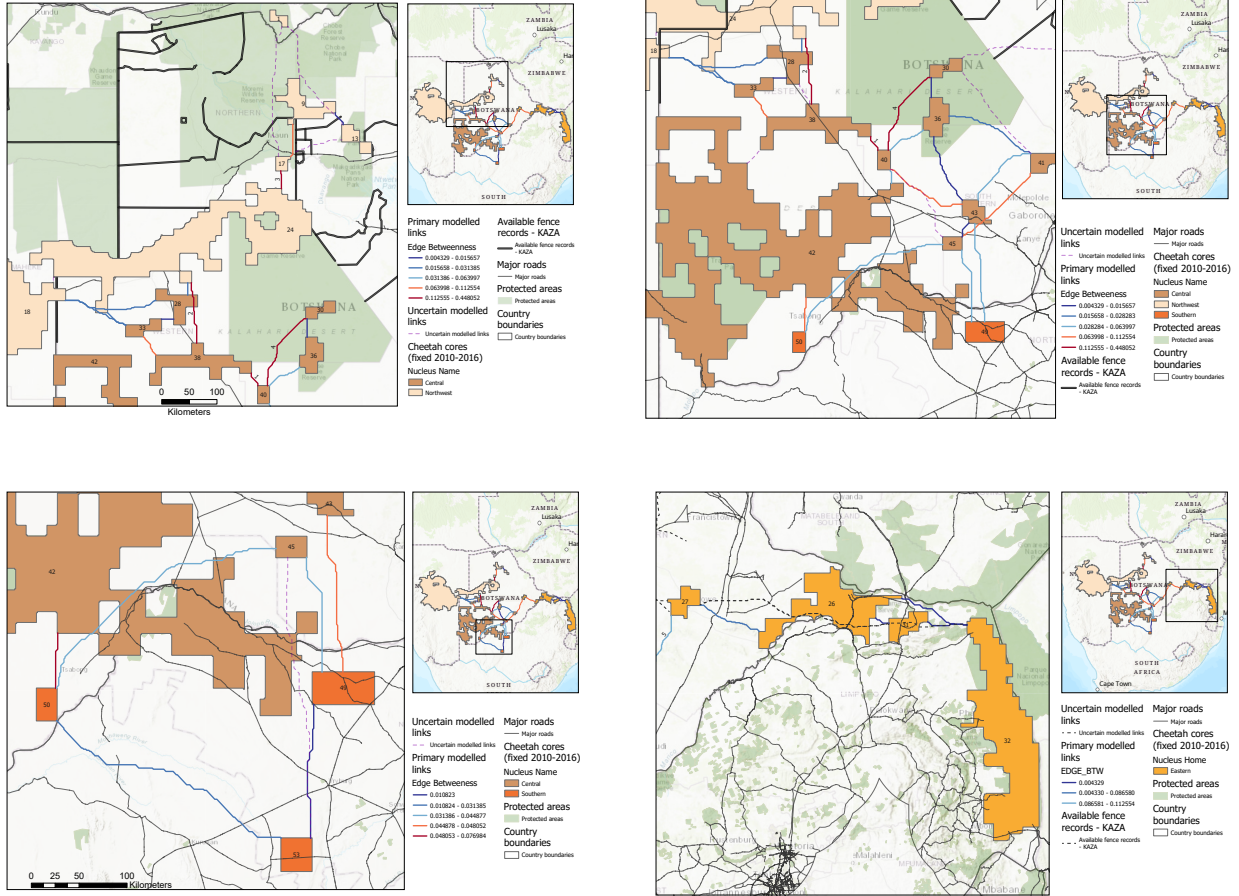


Figure 5: Regional detail maps corresponding to the modeled-link network. These panels support visual inspection of local context and are supplementary to the main network and priority-link figures.